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Lecture – 18 : Other Cryptography Techniques

Faculty & Coordinator : Dr. J Sathish Kumar (JSK)

Department of Computer Science and Engineering

Motilal Nehru National Institute of Technology Allahabad,
Prayagraj-211004

Other Cryptography Techniques

- **DNA Cryptography**
- **Quantum Cryptography**
- **Steganography and**
- **Watermarking**
- ***Acknowledgement:***
  - *Refereed Text book "New Frontiers in Cryptography by Khaled Salah Mohamed*

DNA Cryptography

- DNA cryptography
  - is a promising and rapid emerging field in data security.
  - may bring forward a new hope for unbreakable algorithms.
  - combines cryptology and modern biotechnology.
- Advantages of DNA computing over traditional computing

➢ *Speed*

- Conventional computers have been known to perform approximately 10^8 instructions per second (MIPS).
- Combining DNA strands has been predicted to make computations equivalent to 10^9.

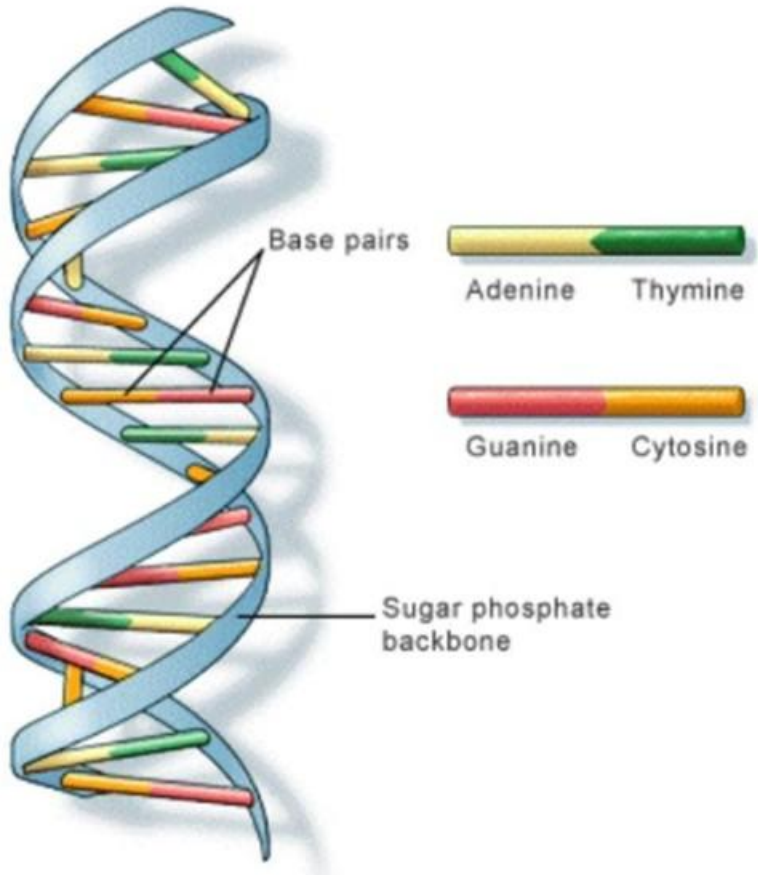
➢ *Storage:*

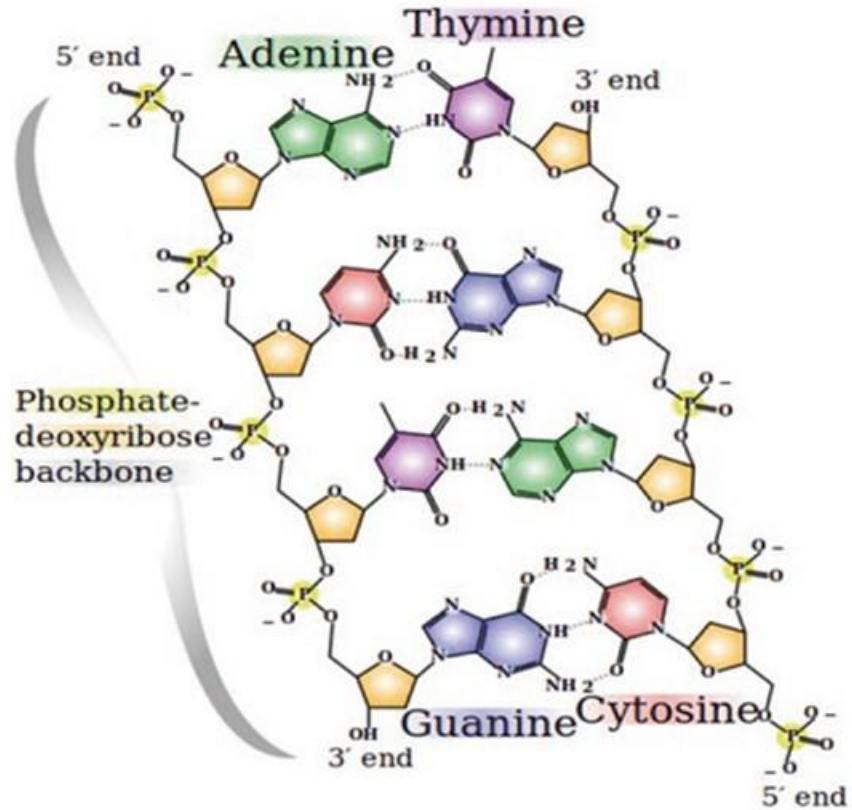
- DNA stores memory at the rate of 1 bit/nm^3, whereas conventional storage media can store 1 bit/10^{12} nm^3.

➢ *Power Requirements:*

- DNA computing does not require power, while computation is taking place.
- The chemical reactions that create the building blocks of DNA take place without any external power source.

DNA Structure





Fundamentals of DNA Computing

- DNA means deoxyribonucleic acid formed using four basic nucleic acids, namely, adenine (A), cytosine (C), guanine (G), and thymine (T)
- The pairs as (A, T) and (C, G) complement each other.
- These alphabets can be easily assigned to binary values *(A-00, C-01, G-10, T-11*).
- By these encoding rules, there are 4! = 24 possible encoding methods.
- However, only eight coding combinations are suitable for the principle of complementarity.
- Because the binary numbers "0" and "1" are complementary, "00" and "11" and "01" and "10" are also complementary

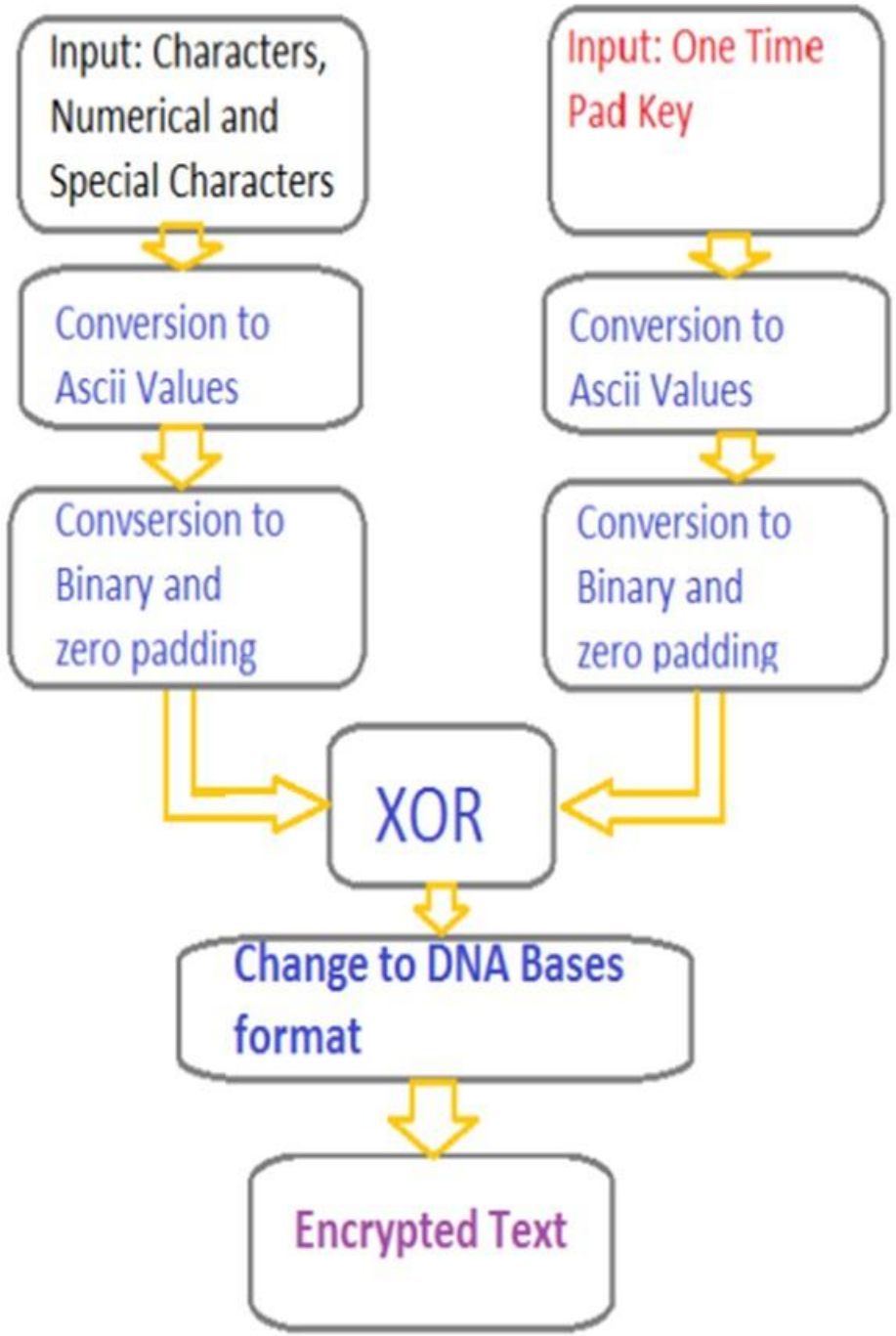
ASCII symbols and their corresponding DNA sequence

S	DNAseq	S	DNAseq	S	DNAseq	S	DNAseq	S	DNAseq	S	DNAseq
' '	ACAT	0	CAAA	@	CCAC	P	TTCA	‘	TCCG	p	GACA
!	AGGT	1	CACC	A	TACT	Q	TTTA	a	GAGC	q	GACT
"	AAAG	2	CCGT	B	TCCT	R	TAGA	b	GTGC	r	GGAT
#	AGAC	3	CGAG	C	TACG	S	TGAG	c	GACG	s	GGTG
$	AAGC	4	CCTT	D	TGCC	T	TAAA	d	GTAA	t	GCTT
%	AACT	5	CCGT	E	TCTA	U	TGAC	e	GTAC	u	GACC
&	AGAA	6	CTGT	F	TAGT	V	TGAG	f	GCCT	v	GACT
'	AATC	7	CTCT	G	TTAA	W	TAAC	g	GCTA	w	GCCC
(	ATTG	8	CCGT	H	TGGC	X	TCCT	h	GAGT	x	GATC
)	AATT	9	CTCA	I	TGTT	Y	TGAA	i	GATG	y	GTCG
*	AATG	:	CTAG	J	TTCC	Z	TAAG	j	GATT	z	GTGA
+	AAGA	;	CCGC	K	TACT	[	TCAT	k	GGGC	{	GGCT
,	AGAG	<	CACA	L	TATG	\\	TAAG	l	GTTG	|	GGTG
-	AAGC	=	CATA	M	TAGT	]	TCCA	m	GTGA	}	GAAC
.	ACAC	>	CTAC	N	TGTC	^	TGTT	n	GACT	~	GATG
/	ACGT	?	CCAG	O	TATT	_	TCCG	o	GCCG	\x7f	GAGT

DNA Cryptography

- To encrypt using DNA, sender generates a DNA encoding table, and receiver generates another table through the same encoding technique and sends a clue to the sender to be able to generate it locally.
- The plaintext to be encoded is divided into two halves equally.
- If the plaintext is not even, we insert random padding.
- One half of the plaintext is converted into DNA sequence using sender-based table, and the other half of the plaintext is converted into DNA sequence using receiver-based table.
- DNA cryptography is a bio-inspired novel technique used for securing end to end communication, where DNA is used as an information carrier.

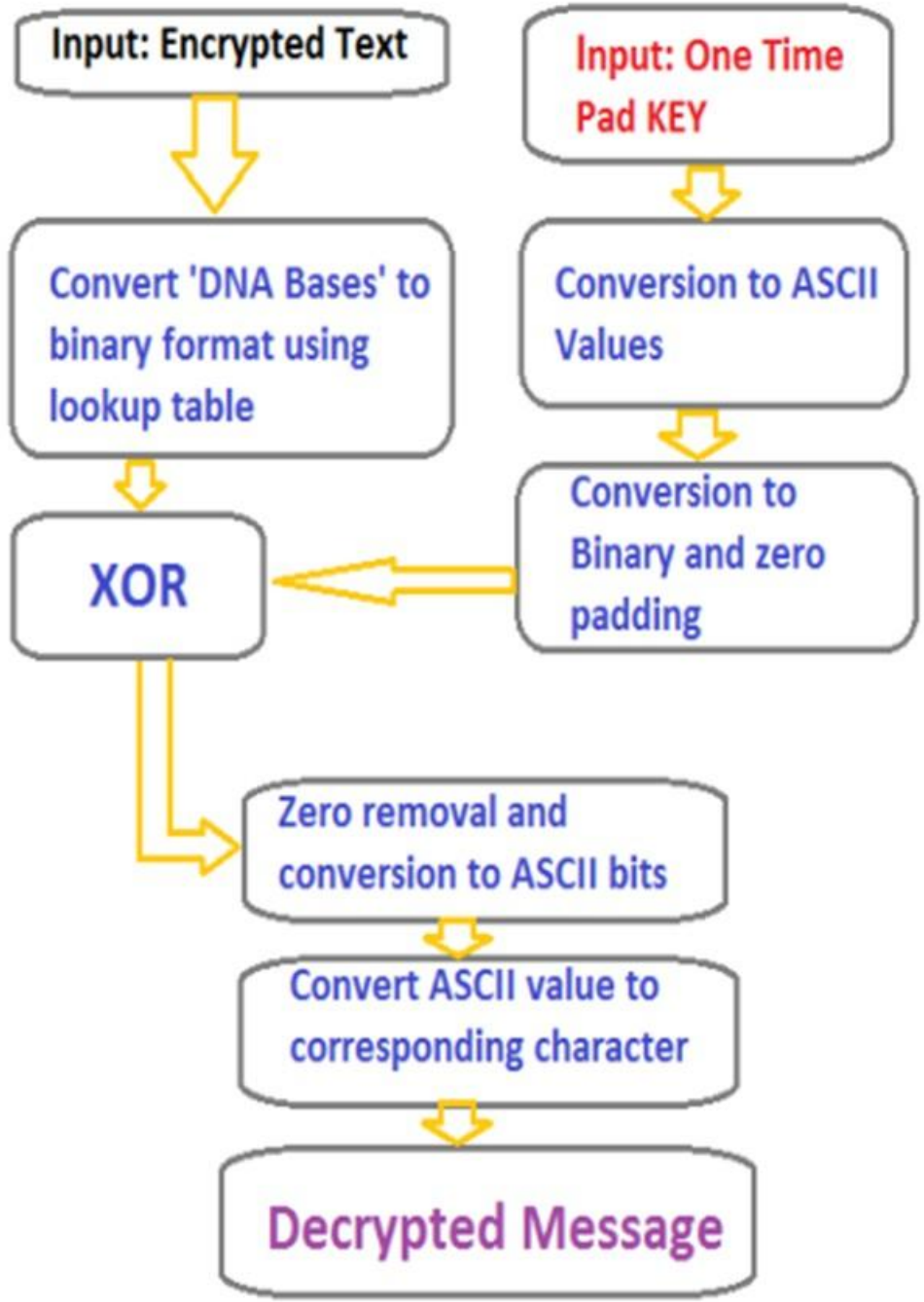
DNA Cryptography Encryption Flow Chart



DNA Cryptography Algorithm

- Convert the plaintext message into an ASCII form and then convert it to 8 bits binary coded form.
- Represent the binary data in the DNA coded form (A-00, C-01, G-10, T-11): *convert encode binary information into DNA strands*.
- Apply complementary rule to the sequence (A → C, C → G, G → T, T → A).
- Convert it back to binary.
- Generate random key and convert it into DND strands then into binary format.
  - The random key has to be a number between 1 and 256.
  - This random key determines the permutation of the four characters A, T, G, and C.
  - For example, when the random key is 1, there is a table for the conversion of ASCII code to nucleotide sequences, and when it is 2, there is another table and so on.
- XORing the key with the data.

DNA Cryptography Decryption Flow Chart



DNA Cryptography Algorithm

- DNA decryption is the reverse operation of DNA encryption.
- The total key space results is approximating to 10^{23}.
- With such huge key space, the reliability and effectiveness of the algorithm are established as the *key* associated is quite unpredictable and resistant against brute force attacks.

Quantum Cryptography

- Quantum cryptography uses physics to develop a cryptosystem completely secure against being compromised without the knowledge of the sender or the receiver of the messages.
- The word *quantum* itself refers to the most fundamental behavior of the smallest particles of matter and energy.
- In 1982, Richard Feynman came up with the idea of quantum computer, a computer that uses the effects of quantum mechanics to its advantage.
- In quantum cryptography, two remote parties can communicate securely by using the laws of quantum physics.
- Quantum cryptography is different from traditional cryptographic systems in that it relies more on physics, rather than mathematics, as a key aspect of its security model.

Properties of Quantum Information

- Quantum cryptography rests on two pillars quantum mechanics :
  - *The Heisenberg uncertainty principle*:
    - It is not possible to measure the quantum state of any system without disturbing that system.
    - Thus, the polarization of a photon or light particle can only be known at the point when it is measured.
  - *The photon polarization principle:*
    - *H*ow light photons can be oriented or polarized in specific directions.
    - Moreover, a photon filter with the correct polarization can only detect a polarized photon or else the photon will be destroyed.
- In quantum cryptography information is transmitted by quantum bit, also called qubit, which is actually a single photon particle.

Quantum Algorithms

- The two basic algorithms of quantum cryptography are Shor's algorithm and the Grover's algorithm
- **Grover's Algorithm**
  - Grover's algorithm is a quantum algorithm that finds with high probability the unique input to a black box function that produces a particular output value.
- **Shor's Algorithm**
  - Shor's algorithm provides a dramatic improvement in the efficiency of factoring large numbers.
  - Thus, Shor's algorithm can be used to attack RSA encryption and related problems.
  - It solves the following problem: Given an integer N, find its prime factors.
- For years, quantum computers have just been research, theory, and proposals. D-Wave is one of the companies that are making quantum computers a reality.

Quantum Cryptography Challenges

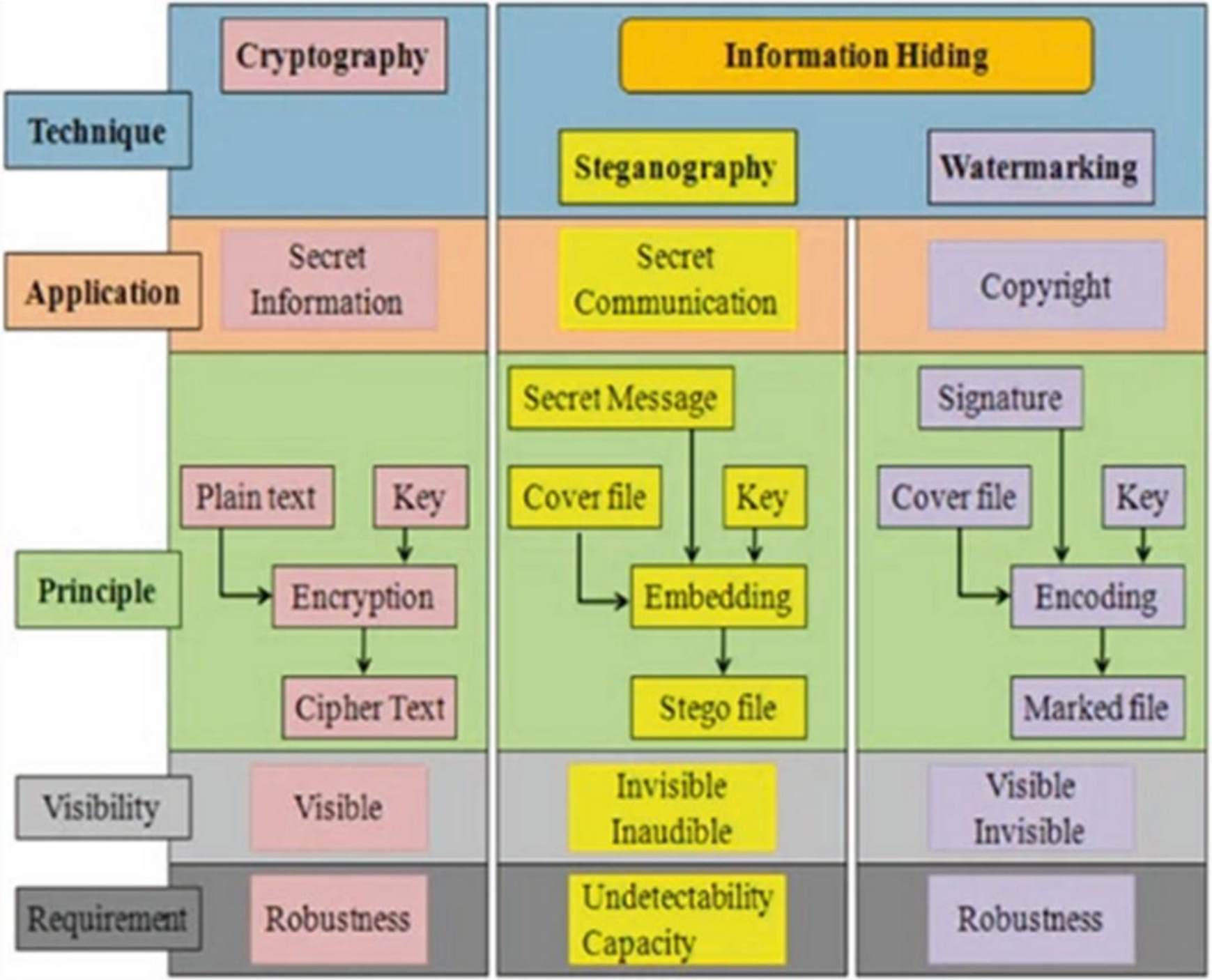
- Expensive: need specialized hardware.
- Complex.
- Difficult to implement over long distance.
- Subjected to de-coherence.
  - Qubits can retain their quantum state for a short period of time.
- Quantum algorithms are mainly probabilistic.
  - This means that in one operation, a quantum computer returns many solutions where only one is the correct.

Steganography

- The word “steganography” comes from the Greek words “stegano” which means “covered” and “graphie” means “writing.”
- Steganography is the art of hiding the existence of a message between sender and intended recipient.
- It hides secret messages in various types of files such as text, images, audio, and video.
- Steganography hides encrypted messages in such a way that no one would even suspect that an encrypted message even exists in the fist place.
- Because there are unused/ redundant data bits in digital media, that changing them will be imperceptible.

Steganography

- There are several techniques to conceal data in the normal files.
- The easiest and simplest method to hide secret information in a digital media is the least significant bit (*LSB*) coding.
- This technique is done by replacing the least significant bit of each sample with a bit of the secret data.
- Steganography is hiding data within data.
- Steganography is often linked to cryptography; however the two are not mutually exclusive.



Comparison between cryptography and information hiding techniques

Comparison between cryptography and information hiding techniques

Comparison	Steganography	Cryptography
Basic	It is known as cover writing	It means secret writing
Goal	Secret communication	Data protection
Structure of the message	Not altered	Altered only of the transmission
Popularity	Less popular	More commonly used
Relies on	Key	No parameters
Supported security principles	Confidentiality and authentication	Confidentiality, data integrity, authentication, and non-repudiation
Techniques	Spatial domain, transform domain, model-based, and ad hoc	Transposition, substitution, stream cipher, block ciphers
Implemented on	Audio, video, image, text	Only on text files
Types of attack	Steganalysis	Cryptanalysis

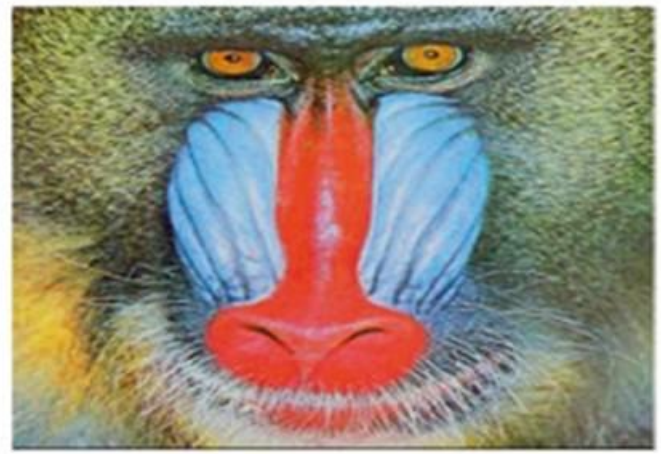
Steganography in Digital Media

- **Text Steganography**
  - The techniques in text steganography are number of tabs, white spaces, and capital letters, just like Morse code is used to achieve information hiding.
  - Various techniques are used to hide the data in the text such as format-based method, random and statistical generation, and linguistic method.
- **Image Steganography**
  - Pixels are the building blocks of an image.
  - Every pixel contains three values (red, green, blue) also known as RGB values.
  - Every RGB value ranges from 0 to 255.
  - In this technique pixel intensities are used to hide the information.
  - The 8 bit and 24 bit images are common.
  - We can even hide an image inside another.

Steganography in Digital Media



(a)



(b)



(c)

(a) Raw image, **(b)** image to be hided, **(c)** stegano-image

Steganography in Digital Media

- **Audio Steganography**
  - Taking audio as carrier for information hiding is called audio steganography.
  - It is used for digital audio formats such as WAVE, MIDI, and AVI MPEG for steganography.
  - The advantage of taking an audio fie as a carrier is that there is a large data redundancy, which means it can be embedded in a large amount of information.
  - Different methods of audio steganography are used such as least significant bit encoding, parity encoding, phase coding, and spread spectrum.

Steganography in Digital Media

- **Video Steganography**
  - It is a technique to hide any type of files or information into digital video format (H.264, Mp4, MPEG, and AVI).
  - The combination of pictures is used as carrier for hidden information.
  - Video steganography is becoming an important research area in various data hiding technologies, which has become a promising tool because not only the security requirement of secret message transmission is becoming stricter but also video is more favored.
  - Two main classes of video steganography are embedding data in uncompressed raw video and compressing it later and embedding data directly into the compressed data stream.

Steganography Techniques

- Image steganography is broadly categorized into two categories, viz., spatial domain image steganography and frequency domain image steganography.
- While in former the message image is simply hidden in the spatial domain of the base image, i.e., no changes are to be done on the cover image for this domain, the latter cover image is converted into frequency domain by using discrete cosine transform, discrete wavelet transform, or any other frequency transformation mechanism, and then the message image is hidden into the coefficients, and further inverse is carried out which is to be transmitted.

Steganography Techniques

- **Least Significant Bit (LSB)**
  - LSB is one the technique of spatial domain methods.
  - Changes in the value of the LSB are imperceptible for human eyes.
  - The image obtained after embedding is almost similar to original image because the change in the LSB of image pixel does not bring too much differences in the image.
  - A digital image consists of a matrix of color and intensity values.
  - In a typical gray-scale image, 8 bits/pixel are used.
  - In a typical full-color image, there are 24 bits/pixel, 8 bits assigned to each color components.
  - Using 8 bit image, the least significant bit (the 8th bit) of some or all of the bytes inside an image is changed to a bit of the secret message.
  - Using 24 bit image, a bit of each of the red, green, and blue color components can be used

Steganography Techniques

- **The Binary Pattern Complexity (BPC)**
  - BPC approach is used to measure the noise factor in the image complexity.
  - The noisy portion is replaced by binary pattern, and it is mapped from the secret data.
- **Discrete Wavelet Transform (DWT)**
  - It is used to transform the image from a spatial domain to the frequency domain.
  - In the process of steganography, DWT identifies the high-frequency and low-frequency information of each pixel of the image.
  - It is a mathematical tool for decomposing an image hierarchically.
  - It is mainly used for processing of non-stationary signals.
  - The wavelet transform is based on small waves, known as wavelets, of different frequency and limited duration.

Steganography Techniques

- **Spread Spectrum (SS)**
  - The concept of spread spectrum is used in this technique.
  - In this method the secret data is spread over a wide frequency bandwidth.
  - The ratio of signal to noise in every frequency band must be so small that it becomes difficult to detect the presence of data.
  - Even if parts of data are removed from several bands, there would be still enough information present in other bands to recover the data.
  - Thus it is difficult to remove the data completely without entirely destroying the cover.

Steganography Techniques

- **Multilevel Steganography (MLS)**
  - In MLS, at least two steganography methods are utilized simultaneously, in such a way that one method (called the upper-level) serves as a carrier for the second one (called the lower-level).
  - This makes the steganogram unreadable.
- **Hybrid Cryptography and Steganography**
  - The data will be encrypted using RSA algorithm or any encryption technique.
  - On the other hand, the encrypted data will be hided in image files or any other fie formats.

Watermarking

- Watermarking is embedding some information within a digital media so that the digital media looks unchanged.
- The hidden information itself is not important by itself, but it says something about the digital media.
- It is a concept closely related to steganography, in that they both hide a message inside a digital signal.
- However, what separates them is their goal.
- Watermarking is used for authenticity (proof of ownership).
- The process of watermarking is divided into two parts:
  - embedding of watermark into host image and extraction of watermark from image
- Watermarking uses the same techniques used by steganography.

Watermarking



(a) (b)

(**a**) Original image, (**b**) watermarked image

Comparison between watermarking and cryptography

Comparison	Cryptography	Watermarking
Basic	It means secret writing	It is known as cover writing
Goal	Data protection	Copyright protection
Structure of the message	Altered only of the transmission	Not altered
Popularity	More commonly used	Less popular
Relies on	No parameters	Key
Supported security principles	Confidentiality, data integrity, authentication, and non-repudiation	Confidentiality and authentication
Techniques	Transposition, substitution, stream cipher, block ciphers	Application-based: Using a universal logo without encryption in the embedding algorithm
Implemented on	Only on text files	Audio, video, image, text
Types of attack	Steganalysis	Cryptanalysis

Watermarking Applications

- Owner Identification:
  - The application of watermarking to which he developed is to identify the owner of any media.
  - Some paper watermark is easily removed by some small exercise of attackers.
  - So the digital watermark was introduced.
  - In that the watermark is an internal part of digital media so that it cannot be easily detected or removed.
  - Embedding a strong watermark is important to prove ownership during conflict resolution process.
- Copy Protection:
  - Illegal copying is also prevented by watermarking with copy protect bit.

Watermarking Applications

- For example, professional photographers tend to watermark digital proofs sent to clients until the client agrees to purchase the photos, at which the original, unaltered images are released.
- This allows the photographer to distribute demos and samples of their work, without actually giving the original compositions.
- *Broadcast Monitoring:*
  - Broadcasting of TV channels and radio news is also monitoring by watermarking.
  - It is generally done with the paid media like sports broadcast or news broadcast.
- *Data Authentication:*
  - Sender embedded the digital watermark with the host data, and it would be extracted at the receivers end and verified.

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